

Software Integration & Maintenance

Fahim Foysal Abak

Lecture, Dept. of CSE

Dhaka International University

COCOMO (Constructive Cost Model)

COCOMO is a **mathematical model** developed by **Barry Boehm** to predict how much **effort and time** a software project will require.

The **COCOMO model** (Constructive Cost Model) is a **software cost estimation model** used to estimate:

- **Effort** (in person-months),
- **Development time**, and
- **Project cost**

based on the **size of the software** (usually measured in *KLOC – thousands of lines of code*).

Categories of projects

Organic

- 📌 Relatively small, simple software projects
- 📌 Small teams with good application experience
- 📌 Similar to previously developed projects
- 📌 Relatively small and require little innovation

Semidetached

- 📌 Intermediate software projects
- 📌 Teams with mixed experience
- 📌 Mix of rigid and less rigid requirements

Embedded

- 📌 Software projects that must be developed within set of tight hardware, software and operational constraints

Categories of projects

P. Characteristics / Development Mode	Size	Innovation	Deadline	Development Environment	KLOC
Organic	Small	Little	Not tight	Stable	2–50
Semidetached	Medium	Medium	Medium	Medium	51–300
Embedded	Large	Greater	Tight	Complex hardware	>300

Stages COCOMO Model

Basic COCOMO

- Uses only **software size (KLOC)**.
- Gives a **rough estimate** of effort and time.
- Suitable in the **early planning stage**.

Intermediate COCOMO

- Uses **size + cost drivers** (e.g., product, hardware, personnel, project attributes).
- Produces **more accurate estimates**.
- Suitable when **more project details** are available.

Detailed COCOMO

- Breaks the project into **development phases** (design, coding, testing, etc.).
- Applies **cost drivers to each phase**.
- Produces the **most accurate estimate**.

Basic COCOMO model

$$Effort = a_1(KLOC)^{a_2} \text{ pm}$$

$$Time = 2.5(Effort)^b \text{ months}$$

$$People = \frac{Effort}{Time} \text{ count}$$

Project Category	a_1	a_2	b
Organic	2.4	1.05	0.38
Semidetached	3.0	1.12	0.35
Embedded	3.6	1.20	0.32

Basic COCOMO model

Consider an **embedded** project with **700KLOC**. Calculate the **effort**, development **time**, average **staff** size, **productivity** to develop the product.

$$\begin{aligned} \textit{Effort} &= a_1(KLOC)^{a_2} \textit{pm} = 3.6(700)^{1.2} \textit{pm} \\ &= 9341.58 \textit{pm} \end{aligned}$$

$$\begin{aligned} \textit{Time} &= 2.5(\textit{Effort})^b \textit{months} = 2.5(9341.58)^{0.32} \textit{months} \\ &= 46.61 \textit{months} \end{aligned}$$

$$\begin{aligned} \textit{Average staff size} &= \frac{\textit{Effort}}{\textit{Time}} \textit{persons} = \frac{9341.58}{46.61} \textit{persons} \\ &= 200.42 \textit{persons} \end{aligned}$$

$$\textit{Productivity} = \frac{KLOC}{\textit{Effort}} = \frac{700}{9341.58} = 0.0749 \textit{KLOC/PM}$$

Intermediate COCOMO model

Basic COCOMO:

Uses only **LOC (size)** and general project type.
Assumes average conditions → **not realistic**.

In reality (Intermediate COCOMO):

Effort depends on **many real-world factors**, not just size.
So Intermediate COCOMO adds **cost drivers** such as:

Cost Drivers — Product Attributes

Cost Drivers	Very Low	Low	Nominal	High	Very High
Required Software Reliability	0.75	0.88	1.00	1.15	1.40
Size of Application Database	—	0.94	1.00	1.08	1.16
Complexity of the Product	0.70	0.85	1.00	1.15	1.30

Intermediate COCOMO model

Cost Drivers — Hardware Attributes

Cost Drivers	Very Low	Low	Nominal	High	Very High
Runtime Performance Constraints	—	—	1.00	1.11	1.30
Memory Constraints	—	—	1.00	1.06	1.21
Volatility of the virtual machine environment	—	0.87	1.00	1.15	1.30
Required turnaround time	—	0.94	1.00	1.07	1.15

Cost Drivers — Personnel Attributes

Cost Drivers	Very Low	Low	Nominal	High	Very High
Analysis Capability	1.46	1.19	1.00	0.86	0.71
Application experience	1.29	1.13	1.00	0.91	0.82
Software engineer capability	1.42	1.17	1.00	0.86	0.70
Virtual machine experience	1.21	1.10	1.00	0.90	—
Programming language experience	1.14	1.07	1.00	0.95	—

Intermediate COCOMO model

Cost Drivers — Project Attributes

Cost Drivers	Very Low	Low	Nominal	High	Very High
Application of software engineering methods	1.24	1.10	1.00	0.91	0.82
Use of software tools	1.24	1.10	1.00	0.91	0.83
Required development schedule	1.23	1.08	1.00	1.04	1.10

Intermediate COCOMO model

$$Effort = a_1(KLOC)^{a_2} \times EAF \text{ } pm$$

$$Time = 2.5(Effort)^b \text{ } months$$

$$People = \frac{Effort}{Time} \text{ } count$$

Project Category	a_1	a_2	b
Organic	3.2	1.05	0.38
Semidetached	3.0	1.12	0.35
Embedded	2.8	1.20	0.32

Intermediate COCOMO model

Consider a project with the estimation of **250KLOC**. Although the size of the application **database** is **nominal** it has **high memory constraints**. The technical team has developers who have **high application experience** but **very low experience in programming**. The **required development schedule for the project is low**. Calculate the **effort**, development **time** and **average staff size** required to develop the product.

$$\begin{aligned} Effort &= a_1(KLOC)^{a_2} \times EAF \text{ pm} \\ &= 3(250)^{1.12} \times (1 \times 1.06 \times 0.91 \times 1.14 \times 1.08) \text{ pm} \\ &= 1727.79 \text{ pm} \end{aligned}$$

$$\begin{aligned} Time &= 2.5(Effort)^b \text{ months} \\ &= 2.5(1727.79)^{0.35} \text{ months} \\ &= 33.97 \text{ months} \end{aligned}$$

$$\begin{aligned} People &= \frac{Effort}{Time} \text{ count} \\ &= \frac{1727.79}{33.97} \text{ persons} \\ &= 50.86 \text{ persons} \end{aligned}$$

Detail COCOMO model

- 📌 All characteristics of intermediate
- 📌 Divided into different modules
- 📌 Apply COCOMO in each modules
- 📌 Sum all efforts

Thank You.